# Pass-Through

C.Conlon

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Grad IO

# What is pass-through and why do we care?

# Pass-Through: What is it?

Lots of cases in economics where we want to know how prices respond to changes in costs:

- Response to cost shocks (oil shock, commodity prices)
- Response to tax changes (incidence/efficiency of taxes)
- Transmission of exchange rate shocks
- Transmission of monetary policy
- Price effects of mergers: converting UPP (a change in opportunity cost) into a change in price
- Double marginalization: how much of a wholesale price change reaches the shelf

Today: what determines pass-through in theory, what the demand models of this course imply for it, and why it is so hard to estimate.

## Hot take

Pass-through is the simplest thing in economics that nobody understands

- Often we are talking about different things
  - IO: mostly levels, $\rho = \frac{\partial p_j}{\partial mc_j}$ or the matrix $\frac{\partial \mathbf{p}}{\partial \mathbf{mc}}$
  - Macro/Trade: mostly logs, $\rho^{\log} = \frac{\partial \log p_j}{\partial \log mc_j}$
  - These are not the same number. With $\mu_j \equiv p_j/mc_j$: $\rho_j^{\log} = \rho_j \cdot \frac{mc_j}{p_j} = \frac{\rho_j}{\mu_j}$.
    Complete pass-through in levels is incomplete pass-through in logs whenever $p > mc$ (Sangani 2026).
  - (Throughout this lecture $\rho$ is the pass-through rate – not the nesting parameter from the demand lectures.)
- One thing we know for sure: constant marginal costs, perfect competition:

$$p = mc \rightarrow \frac{\partial p}{\partial mc} = 1$$

- Everything else depends on two things:
  - Curvature of demand (second derivatives)
  - Nature of competition (conduct)
- Neither is something an own-price elasticity pins down. That is the whole lecture.

# The Empirical Puzzle: Estimates Are All Over the Place

| Shock → price | Units | $\hat{\rho}$ | Source |
|---|---|---|---|
| Exchange rate → US import prices, dollar-priced | logs | 0.25 | Gopinath Itskhoki Rigobon 2010 |
| same, priced in foreign currency | logs | 0.95 | |
| Exchange rate → retail beer | logs | $\approx 0.25$ | Goldberg Hellerstein 2013 |
| Commodity (coffee) → retail | logs | $\approx 0.30$ | Nakamura Zerom 2010 |
| Cigarette taxes | levels | $< 1$ / $\approx 1$ | Harding et al. 2012 / DeCicca et al. 2013 |
| Gasoline taxes | levels | $\approx 1$; holidays $\ll 1$ | Marion Muehlegger 2011; Doyle Samphantharak 2008 |
| Sales taxes | ad val. | $\approx 1$, half $> 1$ | Poterba 1996; Besley Rosen 1999 |
| Alcohol taxes | levels | 1.5 – 2.1 | Young Bielinska-Kwapisz 2002; Kenkel 2005 |

- Trade theory (CES + monopolistic competition) predicts 0 or 1 depending on invoicing currency. The data say 0.25 and 0.95.
- Read the units column: the low numbers are in logs, the numbers near one are in levels. Nakamura–Zerom's 0.30 for coffee is one-for-one in cents (next slides).
- Anything from 0.25 to 2 shows up. Theory says that is exactly what to expect: pass-through is about curvature and conduct, and both vary across markets.
- Many "price elasticity" studies (gasoline, cigarettes, alcohol) assume $\rho = 1$ and regress quantities on tax changes: $\frac{\partial \log q}{\partial \log t} = \frac{\partial \log q}{\partial \log p}$ (!)

# Logs vs. Levels: Sangani (QJE 2026)

Leontief, constant returns: unit cost $c + w$ (commodity $c$, everything else $w$). Two pricing rules:

percentage markup: $p = \mu\,(c + w)$ additive markup: $p = c + w + m$

| | Initial | Change | New | % Change | $\rho^{\log}$ (wrt $mc$) | $\rho$ (levels) |
|---|---|---|---|---|---|---|
| Commodity cost $c$ | \$1 | $+\$0.20$ | \$1.20 | $+20$ | | |
| Marginal cost $c + w$ | \$2 | $+\$0.20$ | \$2.20 | $+10$ | | |
| Price, percentage markup ($\mu = 2$) | \$4 | $+\$0.40$ | \$4.40 | $+10$ | 1.0 | 2.0 |
| Price, additive markup ($m = \$2$) | \$4 | $+\$0.20$ | \$4.20 | $+5$ | 0.5 | 1.0 |

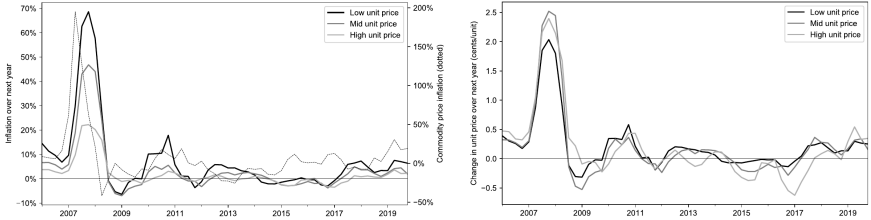
- Regressing $\Delta \log p$ on $\Delta \log c$ estimates $\rho \cdot \frac{c}{p}$: level pass-through times the commodity's share of the price (not of cost). A fixed percentage markup means $\rho = \mu > 1$ in levels.
- Log pass-through is not a stable statistic when markups are additive. For a firm whose whole marginal cost $c$ moves, with $dp = dc$ and $\mu = p/c$, to second order (his Prop. 1):

$$\rho^{\log} = \frac{\Delta \log p}{\Delta \log c} \approx \frac{1}{\mu}\left[1 + \frac{1}{2}\frac{\mu - 1}{\mu}\,\Delta \log c\right]$$

  so a log regression manufactures asymmetry (increases $>$ decreases: "rockets and feathers"), size dependence (big shocks $>$ small ones), and lower pass-through for high-markup firms (large firms, high quality), with no change in behavior.
- Scope: common (industry-wide) shocks to relative input prices. Idiosyncratic shocks are a different experiment (later).

# Sangani (2026): Complete Pass-Through in Levels, Incomplete in Logs

- Perth gasoline (weekly, station level, 2001–2022): long-run pass-through 0.991 (0.038) in levels; 0.899 (0.043) in logs, below gasoline's cost share ($\geq 0.97$).
- Interact with station markups: in levels the interaction is $\approx 0$; in logs it is $-0.010$ to $-0.015$ per point of net markup (OLS and IV), what additive markups predict ($\approx -0.01$). As markups $\to 0$, log pass-through $\to$ the cost share.
- Food (BLS average prices, 12 months), logs / levels: coffee 0.47 / 0.95, rice 0.31 / 0.88, flour 0.24 / 0.87, OJ 0.33 / 0.97, beef 0.41 / 0.90; sugar 0.37 / 0.69 is the exception.
- Manufacturing (NBER-CES, 459 industries): $\Delta \log p$ on $\Delta \log c \times \frac{\text{input cost}}{\text{sales}}$ gives $\rho \approx 0.95$–$1.04$ (eq. (9) of the paper).




(A) Inflation (in percentages)

(B) Price change (in levels)

Rice in NielsenIQ, by tercile of unit price. 2008 run-up: cheap rice inflates 70%, expensive rice under 25% (left); all three rise by the same cents per unit (right).

# One product, one price

# Perfect Competition: Jenkin (1872), Alfred Marshall (1890)

$$D(p) = S(p - t)$$

$$\rho = \frac{dp}{dt} = \frac{1}{1 + \frac{\varepsilon_d}{\varepsilon_s}}$$

- Where $\varepsilon_d$ is demand elasticity and $\varepsilon_s$ is supply elasticity respectively.
- $\rho \in [0, 1]$ as long demand slopes down and supply slopes up!
- Constant MC implies that $\varepsilon_s \to \infty$ and $\rho \to 1$
- Incidence $I = \frac{\frac{dCS}{dt}}{\frac{dPS}{dt}} = \frac{\rho}{1-\rho}$ (higher pass-through, more borne by consumers)

So far so good. Nothing about curvature yet: with perfect competition only slopes matter.

# Monopoly (Weyl and Fabinger, JPE 2013)

Start with $MR = MC$ and differentiate with respect to the tax $t$:

$$p(q) + p'(q)\,q = mc(q) + t \quad \Rightarrow \quad \frac{dq}{dt} = \frac{1}{mr' - mc'} \quad \Rightarrow \quad \rho = \frac{dp}{dt} = p'\frac{dq}{dt} = \frac{p'}{mr' - mc'}$$

Define the markup term $ms(q) \equiv -p'(q)\,q$ and the elasticities $\epsilon_D = -\frac{p}{p'q}$, $\epsilon_S = \frac{mc}{mc'q}$, $\epsilon_{ms} = \frac{ms}{ms'q}$. Using the Lerner index $\frac{mc}{p} = \frac{\epsilon_D - 1}{\epsilon_D}$:

$$\boxed{\rho = \frac{1}{1 + \frac{\epsilon_D - 1}{\epsilon_S} + \frac{1}{\epsilon_{ms}}}} \qquad \text{constant MC: } \rho = \frac{1}{1 + 1/\epsilon_{ms}}, \qquad \frac{1}{\epsilon_{ms}} = 1 + \frac{p''q}{p'}$$

- $\epsilon_D > 1$ for a monopolist (except for my MBA students).
- $1/\epsilon_{ms}$ is the curvature of (inverse) demand, and $(\log D)'' = -\frac{1}{\epsilon_{ms}}\frac{1}{ms^2}$, so its sign is the sign of the curvature of log demand:
  - log-concave demand $\Rightarrow 1/\epsilon_{ms} > 0 \Rightarrow \rho < 1$ (linear demand: $\rho = 1/2$)
  - log-linear (exponential) demand $\Rightarrow \rho = 1$
  - log-convex demand $\Rightarrow \rho > 1$ (constant elasticity)
- Bulow and Pfleiderer (1983): with constant MC a monopolist's pass-through is entirely about curvature.

# Canonical Demand Curves (single product, constant MC)

| Demand | $\rho = dp/dc$ | log $D$ | Markup |
|---|---|---|---|
| Perfect competition | $1$ | – | $0$ |
| Linear $q = a - bp$ | $1/2$ | concave | $(p - c) = \frac{a-bc}{2b}$ |
| Exponential $q = Ae^{-\lambda p}$ | $1$ | linear | $p - c = 1/\lambda$ (constant) |
| Constant elasticity $q = Ap^{-\epsilon}$ | $\frac{\epsilon}{\epsilon-1} > 1$ | convex | $p = \frac{\epsilon}{\epsilon-1} c$ (constant Lerner) |
| Logit, single-product firm, rivals fixed | $1 - \sigma_j$ | concave | $p_j - c_j = \frac{1}{\alpha(1-\sigma_j)}$ |

- Linear, exponential and constant-elasticity are the $\rho = \frac{1}{2}, 1, > 1$ members of the Bulow–Pfleiderer (1983) constant pass-through family; the puzzle slide lives between the linear and CES rows.
- A constant absolute markup (exponential) gives $\rho = 1$; a constant percentage markup (CES) gives $\rho > 1$ because the markup scales with cost. Hence CES trade models over-shift unless pricing-to-market or invoicing is added. These are the two benchmarks on the logs-vs-levels slide: complete in levels vs. complete in logs.
- Logit sits below one and gets closer to one as the product gets smaller. Derivation in two slides.

## Conduct and Incidence (Weyl and Fabinger 2013)

Generalize to symmetric imperfect competition with a conduct parameter $\theta$: $\theta = 0$ (PC), $\theta = 1$ (monopoly), $\theta = 1/N$ (symmetric Cournot):

$$\rho = \frac{1}{1 + \frac{\theta}{\epsilon_\theta} + \frac{\epsilon_D - \theta}{\epsilon_S} + \frac{\theta}{\epsilon_{ms}}} \qquad \text{with } \epsilon_\theta \equiv \frac{\theta}{\theta' q} \text{ (how conduct itself changes with quantity).}$$

Incidence follows from pass-through and conduct alone (their "Principle of Incidence"):

$$\frac{1}{I} = \frac{1}{\rho} - (1 - \theta) \iff I = \frac{\rho}{1 - (1 - \theta)\rho} : \qquad I_{PC} = \frac{\rho}{1 - \rho}, \qquad I_{\text{monopoly}} = \rho.$$

- Monopoly: producer surplus falls by exactly $q$ per dollar of tax (envelope theorem), consumer surplus by $\rho\, q$. So $\rho > 1$ means consumers bear more than the whole tax.
- Two sufficient statistics for incidence, $(\rho, \theta)$, neither of which is an elasticity.
- Not sure how I feel about the revival of the conduct parameter (more in the conduct lecture).

# Elasticity vs. Curvature: Mrázová and Neary (AER 2017)

Write everything in terms of inverse demand $p(x)$. Two local statistics at any point:

$$\varepsilon(x) \equiv -\frac{p(x)}{x\,p'(x)} > 0 \quad \text{(elasticity)} \qquad \kappa(x) \equiv -\frac{x\,p''(x)}{p'(x)} \quad \text{(convexity; MN call it } \rho\text{)}$$

Monopoly, constant MC: $\dfrac{dp}{dc} = \dfrac{1}{2-\kappa} \quad \Rightarrow \quad \dfrac{dp}{dc} - 1 = \dfrac{\kappa - 1}{2-\kappa}$

- Second-order condition: $\kappa < 2$. Over-shifting ($\rho > 1$) iff $\kappa > 1$: demand more convex than exponential.
- Superconvex demand ($\kappa > 1 + 1/\varepsilon$, more convex than CES at the same elasticity): the proportional markup $p/c$ *rises* with cost. CES is the boundary.
- The demand manifold: any demand function traces a curve in $(\kappa, \varepsilon)$ space. The elasticity sets the markup, the convexity sets the pass-through, and knowing one tells you nothing about the other.
- Our problem in one line: demand estimation targets $\varepsilon$; cost-change counterfactuals need $\kappa$, which the functional form supplies for free.

# Mrázová and Neary (2017): The Space of Elasticity and Convexity

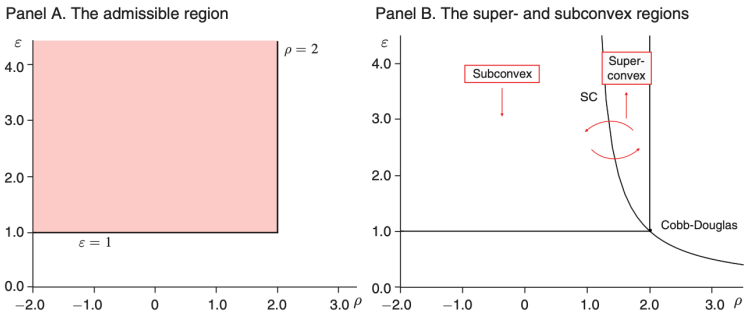



FIGURE 1. THE SPACE OF ELASTICITY AND CONVEXITY

Panel A: the admissible region ($\varepsilon > 1$, $\kappa < 2$). Panel B: the SC curve is the CES family ($\varepsilon$ constant); below it demand is subconvex (proportional markups fall with cost), above it superconvex. Cobb–Douglas is the corner.

# Discrete choice

# Logit Pass-Through in Two Lines

Single-product firm, plain logit, rivals' prices held fixed. The FOC is $p_j = c_j + \eta_j(p_j)$ with

$$\eta_j = \frac{1}{\alpha(1-\sigma_j)}, \qquad \frac{\partial \eta_j}{\partial p_j} = \frac{1}{\alpha}\frac{1}{(1-\sigma_j)^2}\frac{\partial \sigma_j}{\partial p_j} = -\frac{\sigma_j}{1-\sigma_j} \qquad \Rightarrow \qquad \rho_j = \frac{1}{1-\partial \eta_j/\partial p_j} = \boxed{1-\sigma_j}$$

- Logit demand is log-concave in own price: $(\log \sigma_j)'' = -\alpha^2 \sigma_j(1-\sigma_j) < 0$, so $\rho_j < 1$ always.
- The markup $\frac{1}{\alpha(1-\sigma_j)}$ *falls* as the price rises (share shrinks), which is what absorbs part of the cost increase. A niche product passes through almost everything; a dominant product very little.
- Note what is not in the formula: $\alpha$. The price coefficient sets the level of markups and elasticities; the curvature, and hence pass-through, is pinned down by the share alone. This is the second-derivative analogue of IIA.
- With rivals responding (strategic complements) own pass-through is higher and cross pass-through is positive: that needs the matrix (next section). The $1-\sigma_j$ is its "rivals fixed" diagonal. A cost shock common to every product is a third experiment, with its own answer (after the matrix).

# Mixed Logit: Where Curvature Comes From (Miravete, Seim, Thurk 2023)

Aggregate demand is a mixture, $\sigma_j(\mathbf{p}) = \sum_i w_i\, \sigma_{ij}(\mathbf{p})$, and mixtures of log-concave functions need not be log-concave. Let price enter utility through a price-response subfunction $f_i(y_i, p_j)$ of income $y_i$ and price:

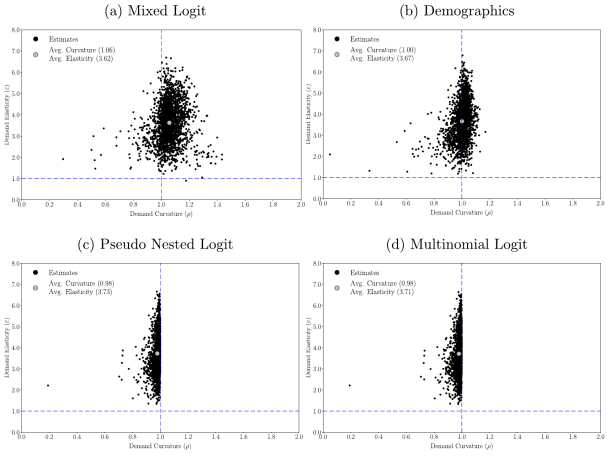
$$u_{ij} = x_j\beta_i + f_i(y_i, p_j) + \xi_j + \epsilon_{ij}, \qquad f'_{ij} \equiv \frac{\partial f_i}{\partial p_j}, \quad f''_{ij} \equiv \frac{\partial^2 f_i}{\partial p_j^2}.$$

Writing $\omega_{ij} \equiv \sigma_{ij}(1-\sigma_{ij})$, own elasticity and curvature (MST call $\kappa$ "$\rho$"; it is Mrázová–Neary's $\kappa$):

$$\varepsilon_j = -\frac{p_j}{\sigma_j}\sum_i w_i\, f'_{ij}\,\omega_{ij}, \qquad \kappa_j = \frac{\sigma_j \sum_i w_i \left[f''_{ij}\,\omega_{ij} + (f'_{ij})^2\,\omega_{ij}(1-2\sigma_{ij})\right]}{\left[\sum_i w_i\, f'_{ij}\,\omega_{ij}\right]^2}, \qquad \left.\frac{dp_j}{dc_j}\right|_{\text{rivals fixed}} = \frac{1}{2-\kappa_j}$$

- Check: one type, $f = -\alpha p_j$ gives $\kappa_j = \frac{1-2\sigma_j}{1-\sigma_j}$ and $\frac{1}{2-\kappa_j} = 1-\sigma_j$, the logit slide.
- Two sources of curvature. (1) Nonlinear price, $f'' \neq 0$: $\alpha \log(y_i - p_j)$ (BLP 1995), Box–Cox in $y_i - p_j$ (MST's companion WP). (2) Heterogeneous price sensitivity $f'_{ij} = -\alpha_i$: raise $p_j$ and the price-sensitive types leave first, so residual demand is less elastic than the average and $\kappa_j > 1$ becomes possible. Random coefficients on *characteristics* with a common $\alpha$ pull curvature down; in MST's estimates logit and nested logit never get past $\kappa = 1$, so they under-shift by construction.
- Quasi-linear $f_i = \alpha_i(y_i - p_j)$ kills (1), so curvature comes only from the distribution you assumed for $\alpha_i$, which prices and shares barely discipline. Normal, lognormal, or two-point $\alpha_i$ can fit the same elasticities and imply different pass-through. ($f'' = 0$ is also what makes demand shift invariant in the common-shocks section.)

# Miravete, Seim, Thurk (IJIO 2023): Same Elasticities, Different Curvature




- Nevo (2000) cereal data, four specifications: (a) mixed logit, (b) $\alpha_i$ and $\beta_i$ from demographics only, (c) random coefficients on characteristics only ("pseudo nested logit"), (d) plain logit. Each dot is a product's $(\kappa, \varepsilon)$; horizontal axis is curvature.
- Average elasticities 3.62–3.73 and Lerner indices ($\approx 0.3$) are nearly identical across models. The curvature distributions are not: (c) and (d) are capped at $\kappa = 1$; (a) spreads on both sides.
- 10% marginal-cost increase (multiproduct Bertrand): mixed-logit pass-through centers near 1.03, with a wide spread; plain and nested logit are all below 1.
- Merger of the two largest firms: mixed logit predicts roughly twice the price increase (median $\approx 11\%$ vs. $\approx 6$–$7\%$) and needs larger synergies to offset ($\approx 27\%$ vs. $\approx 15\%$). (Read off box plots.)

Practical advice: (a) report the per-product $(\varepsilon, \kappa)$ scatter, not the average elasticity, and the implied pass-through next to it; (b) if you have cost or tax variation, use it as a supply moment (as Conlon Rao do in the BLP lecture); (c) if your counterfactual is a cost change, the mixing distribution and the price subfunction are doing the work, so vary them.

# Many products: the pass-through matrix

# Multiproduct Bertrand: Pass-Through Is a Matrix

Recall the FOC in markup form from the merger simulation lecture, $\Omega(\mathbf{p}) \equiv \mathcal{H} \odot \Delta(\mathbf{p})$:

$$f(\mathbf{p}, \mathbf{mc}) \equiv \mathbf{p} - \mathbf{mc} - \underbrace{\Omega(\mathbf{p})^{-1}\boldsymbol{\sigma}(\mathbf{p})}_{\boldsymbol{\eta}(\mathbf{p})} = 0.$$

The implicit function theorem, with $\partial f / \partial \mathbf{mc} = -\mathbb{I}_J$ because costs enter additively:

$$\mathcal{P} \equiv \frac{\partial \mathbf{p}}{\partial \mathbf{mc}} = -\left(\frac{\partial f}{\partial \mathbf{p}}\right)^{-1} \frac{\partial f}{\partial \mathbf{mc}} = \left(\mathbb{I}_J - \frac{\partial \boldsymbol{\eta}}{\partial \mathbf{p}}\right)^{-1}, \qquad \mathcal{P}_{(j,k)} = \frac{\partial p_j}{\partial mc_k}.$$

- Diagonal: own-cost pass-through, now including rivals' equilibrium responses. Off-diagonal: a rival's cost shock moves my price (positive under strategic complements).
- Single-product logit: $\partial \eta_j / \partial p_j = -\sigma_j/(1-\sigma_j)$ on the diagonal; ignoring the off-diagonal terms gives back $1 - \sigma_j$.
- Multiproduct firms: $\eta_j$ depends on the margins of all sister products, so a cost shock to $k$ moves $p_j$ with no rival response at all.
- Everything hard is in $\partial \boldsymbol{\eta} / \partial \mathbf{p}$: the derivative of a markup is a second derivative of demand.

## How do we get the pass-through matrix?

Two ways:

1. Numerically: perturb the marginal cost of each product $k$ in turn and re-solve for equilibrium prices

$$\mathbf{p} = \mathbf{mc} + e_k + \Omega(\mathbf{p})^{-1}\boldsymbol{\sigma}(\mathbf{p})$$

   (Morrow Skerlos (2011) fixed point, $J$ solves, finite differences; fine for small $J$.)

2. Analytic: differentiate the FOC and apply the implicit function theorem (next slide).
   See Jaffe Weyl (AEJM 2013), Miller Weinberg (2017, Appendix E), Conlon Rao (2025).
   This is what PyBLP does with `results.compute_passthrough`, building the full $J \times J \times J$ demand Hessian (slow, and $J^3$ memory). JaxBLP contracts that tensor with the markups analytically first and never forms it ($J^2 I$ flops; 0.4s vs. 3.7s and 31GB on a 636-product market).

## Multivariate IFT: The Hard Part

Differentiate the FOC with respect to $p_l$ to get column $l$ of $\partial f / \partial \mathbf{p}$:

$$\frac{\partial f(\mathbf{p}, \mathbf{mc})}{\partial p_l} \equiv e_l + \Omega^{-1}(\mathbf{p}) \left[ \mathcal{H} \odot \frac{\partial \Delta(\mathbf{p})}{\partial p_l} \right] \Omega^{-1}(\mathbf{p}) \, \boldsymbol{\sigma}(\mathbf{p}) - \Omega^{-1}(\mathbf{p}) \, \frac{\partial \boldsymbol{\sigma}(\mathbf{p})}{\partial p_l}. \tag{1}$$

The complicated piece is the demand Hessian: a $J \times J \times J$ tensor with elements $(j, k, l)$, $\frac{\partial^2 \sigma_j}{\partial p_k \partial p_l}$, since $\frac{\partial \Delta(\mathbf{p})}{\partial p_l} = -\frac{\partial^2 \boldsymbol{\sigma}}{\partial \mathbf{p} \partial p_l}$.

- Compactly, with markups $\boldsymbol{\eta} = \Omega^{-1} \boldsymbol{\sigma}$: $\frac{\partial f}{\partial \mathbf{p}} = \mathbb{I}_J + \Omega^{-1} \left[ T - \frac{\partial \boldsymbol{\sigma}}{\partial \mathbf{p}} \right]$ with $T_{(j,l)} = \sum_m \frac{\partial \Omega_{(j,m)}}{\partial p_l} \eta_m$. The tensor only ever appears contracted with the markups; computing $T$ directly is what makes the closed form cheap.
- Orientation (as in the pricing lecture): with product-specific $\alpha_{ik}$, read $\Omega = \mathcal{H} \odot \Delta'$.
- For (mixed) logit the Hessian is a sum of products of $\sigma_{ij}$'s and $\alpha_i$'s: the "curvature from mixing" slide, in matrix form.

# Application: From UPP to Price Effects (Jaffe Weyl 2013)

Recall UPP from the antitrust deck: $UPP_j = \Delta c_j + \sum_{k \in \mathcal{J}_g} (p_k - c_k) \cdot D_{jk}(\mathbf{p})$ is a change in opportunity cost. But how much do we expect prices to rise?

- Treat the merger as a cost shock and pass it through: $\Delta \mathbf{p} \approx \mathcal{P} \cdot \mathbf{UPP}$. The first-order approach to merger analysis.
- That needs the whole $J \times J$ matrix $\mathcal{P}$, which is $J^2$ numbers (again). So people restrict it: diagonal, or $\mathcal{P} = 0.8 \cdot \mathbb{I}_J$, or just the merging firms' own-cost pass-through.
- Miller, Remer, Ryan, Sheu (2016) simulate mergers under several demand systems and ask how good the approximation is with the full matrix, the diagonal, or only the merging firms' rows (next slide).
- Which approximations work depends on demand curvature, and curvature is exactly what the approximation is trying to avoid estimating.

# Miller, Remer, Ryan, Sheu (2016): How Good Is the First-Order Approximation?

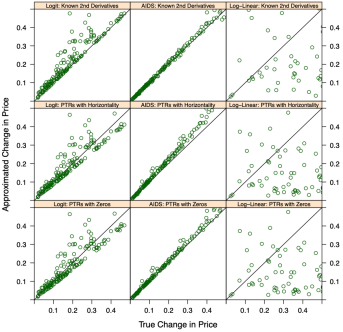



Figure 2: Prediction Error with Complete Information.

Notes: The figure provides scatter-plots of approximation against the true price effect for logit demand, the AIDS and log-linear demand. The case of linear demand is omitted because approximation is exact in that setting. Approximations are calculated alternately based on full knowledge of the second-order properties of demand in neighborhood of pre-merger equilibrium ("Known 2nd Derivatives"); based on full knowledge of pre-merger cost pass-through and the horizontality assumption ("PTRs with Horizontality"); and based on full knowledge of pre-merger cost pass-through and the assumption that derivatives of the form $\partial^2 Q_m/\partial P_j \partial P_k$ equal zero ("PTRs with Zeros").

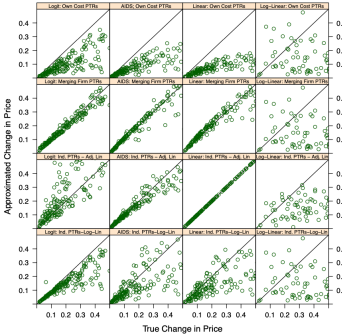



Figure 3: Prediction Error with Incomplete Information

Notes: The figure provides scatter-plots of approximations against the true price effect for logit demand, the AIDS, linear demand and log-linear demand. Four informational scenarios are considered: pre-merger cost pass-through that is available only for own costs ("Own Cost PTRs"); pre-merger cost pass-through that is available only for the merging firms ("Merging Firms' PTRs"); industry cost pass-through that is apportioned using the adjusted-linear method ("Ind. PTRs–Adj.–Lin."); and industry cost pass-through that is apportioned using the log-linear method ("Ind. PTRs–Log–Lin").

Left (complete information): with the full pass-through matrix the approximation is close to exact for AIDS and acceptable for logit; log-linear (CES-like) demand is hopeless because the price effect is far from linear in the cost shock. Right (incomplete information): using only own-cost pass-through understates every price effect; the merging firms' rows do most of the work.

# Common shocks: shift vs. scale invariance

# Which Demand Systems Pass a Common Shock Through in Levels? (Sangani 2026)

A common shock raises every inside good's cost by $dc$ (outside prices fixed). Its price response is the row sum $\mathcal{P}\mathbf{1}$, not the diagonal. Let $q_j(\mathbf{p})$ be the quantity demanded of $j$, holding outside prices and income fixed; in discrete choice $q_j = M\sigma_j$ with market size $M$.

- Scale invariance: $q_j(\lambda\mathbf{p}) = \lambda^{\varphi_j} q_j(\mathbf{p})$ with $\varphi_j$ not depending on $p_j$. A proportional shift leaves every elasticity unchanged $\Rightarrow$ complete in logs, so $\rho = \mu > 1$ in levels (Prop. 4).
  All homothetic demand: CES, Kimball, HSA, nested CES (Atkeson–Burstein), nested Kimball (Amiti–Itskhoki–Konings). Their variable markups only show up for idiosyncratic shocks.
- Shift invariance: $q_j(\mathbf{p} + \lambda\mathbf{1}) = (1 + \lambda\psi_j)\, q_j(\mathbf{p})$ with $\psi_j$ not depending on $p_j$. A uniform shift rescales each residual demand curve, leaving the additive markup $q_j/|\partial q_j/\partial p_j|$ unchanged $\Rightarrow$ complete in levels (Prop. 5). No demand system is both (Prop. 6).
- Leading case, $\psi_j = 0$: quasi-linear random utility $u_{ij} = \delta_{ij} - p_j$ where every consumer buys one of the inside goods (McFadden's translation invariance; Anderson, de Palma, Thisse 1992). That covers logit, nested logit, mixed logit with any mixing distribution, probit, Hotelling, and Salop. (Symmetric logit and Salop: Anderson, de Palma, Kreider 2001a,b already show the markup does not depend on cost, so a unit tax is fully passed on.)

In our notation: if $\boldsymbol{\sigma}(\mathbf{p} + \lambda\mathbf{1}) = \boldsymbol{\sigma}(\mathbf{p})$, then $\Delta$ and $\boldsymbol{\eta} = \Omega^{-1}\boldsymbol{\sigma}$ depend only on price *differences*, so

$$\frac{\partial \boldsymbol{\eta}}{\partial \mathbf{p}}\,\mathbf{1} = \mathbf{0} \quad \Rightarrow \quad \left(\mathbb{I}_J - \frac{\partial \boldsymbol{\eta}}{\partial \mathbf{p}}\right)\mathbf{1} = \mathbf{1} \quad \Rightarrow \quad \boxed{\mathcal{P}\,\mathbf{1} = \mathbf{1}}$$

for any ownership matrix $\mathcal{H}$. The diagonal of $\mathcal{P}$ can be anything the curvature says; the row sums cannot.

# The Outside Good Is What Breaks It

An outside good at a fixed price breaks translation invariance: a common increase pushes consumers out of the market. (So do income effects, $\alpha \log(y_i - p_j)$ as in BLP 1995 and Petrin 2002.) Symmetric Bertrand (Anderson, de Palma, Kreider 2001b, eq. 9), with $\varepsilon_{dd}$ the own-price elasticity, $\varepsilon_{DD}$ the elasticity when all prices move together, and $\varepsilon_m$ the elasticity of the $dd$ slope along $DD$:

$$\rho^{\text{common}} = \frac{\varepsilon_{dd}}{\varepsilon_{dd} + \varepsilon_{DD} - \varepsilon_m} \qquad (\text{covered market: } \varepsilon_{DD} = \varepsilon_m = 0 \Rightarrow 1).$$

Symmetric single-product logit with inside shares $\sigma_j$ and outside share $\sigma_0$ (derived here): $\varepsilon_{dd} = -\alpha p(1-\sigma_j)$, $\varepsilon_{DD} = -\alpha p\,\sigma_0$, $\varepsilon_m = -\alpha p\,\sigma_0 \frac{1-2\sigma_j}{1-\sigma_j}$, so

$$\rho^{\text{common}} = \frac{1}{1 + \frac{\sigma_j}{1-\sigma_j}\, D_{j0}} \quad \text{vs.} \quad 1 - \sigma_j = \frac{1}{1 + \frac{\sigma_j}{1-\sigma_j}}, \qquad D_{j0} = \frac{\sigma_0}{1-\sigma_j} \text{ (logit diversion to the outside good).}$$

- No diversion to the outside good, or atomistic firms ($\sigma_j \to 0$; Sangani's $p_j = c_j + p_0/b_j$) $\Rightarrow$ complete in levels.
- Mixed logit with an outside good can give anything. Numerical example: 4 products, three types $\alpha_i \in \{0.5, 1.5, 4\}$, $\sigma_0 \approx 0.72$. Row sums of $\mathcal{P}$ are 1.76–1.89, because the price-sensitive types leave first. Drop the outside good, keep the same $(\delta, \alpha_i)$: exactly 1.
- Own and common pass-through use different curvature ($dd$ vs. $DD$). Nakamura–Zerom's residual superelasticity, in Kimball demand, still gives complete *log* common pass-through.
- For estimation: level pass-through of industry-wide shocks is a cheap moment that disciplines what prices and shares do not, the outside-good margin and the mixing distribution.

# Why Logs vs. Levels Matters Beyond Measurement (Sangani 2026)

- Margins and entry. Percentage markups plus inelastic industry demand mean a cost increase raises profit per unit, so operating margins or entry must rise. Data: gross margins fall with input prices (elasticity $-0.26$ to $-0.29$ for retail gasoline, $-0.19$ for NBER-CES manufacturing). Operating margins and establishment counts do not move. Additive markups predict exactly this.
- Dispersion. Additive markups mean a common cost increase compresses the dispersion of log prices, percentage markups, and TFPR / labor-productivity dispersion, with *no* reallocation of quantities (rice, flour, coffee). Under non-homothetic demand, falling markup dispersion is not evidence of less misallocation.
- Measured markups. Percentage markups and Lerner indices fall mechanically when input prices rise, with no change in conduct. Keep this in mind when reading markup time series.
- Macro. Input–output model of the US (402 industries, Calvo). With CES (percentage) markups, CPI inflation has a standard deviation of 2.5% vs. 1.1–1.3% in the data. Calling labor a variable cost lowers it, but implies markups of 1.17–1.31. Logit (additive) markups give 1.3% and match Känzig's oil-shock response of the CPI (peak 0.43 vs. 0.45).
- Okun (1981): firms pass materials through "on a dollars-and-cents basis" and labor "with a percentage markup." In logit with $\alpha$ measured against the wage, the additive markup is $w/b$, so wage shocks move it.

# Vertical chains

# Double Marginalization: Villas-Boas (ReStud 2007), Miller Weinberg (Ecma 2017)

Most retail prices are set by a retailer who buys from a manufacturer with market power. Retailer and wholesaler FOCs:

$$\mathbf{p^r} = \underbrace{\mathbf{p^w} + \mathbf{c^r}}_{\mathbf{mc^r}} + (\mathcal{H}_r \odot \Delta_r(\mathbf{p^r}))^{-1}\boldsymbol{\sigma}(\mathbf{p^r})$$

$$\mathbf{p^w} = \mathbf{mc^w} + \left(\mathcal{H}_w \odot \left(\left[\frac{\partial \mathbf{p^r}}{\partial \mathbf{p^w}}\right]' \cdot \Delta_r(\mathbf{p^r})\right)\right)^{-1} \boldsymbol{\sigma}(\mathbf{p^r})$$

- $\Delta_r$ is the matrix of (retail) demand derivatives with entries $-\frac{\partial \sigma_j}{\partial p_k^r}$ (as in the merger simulation lecture).
- $\mathcal{H}_r, \mathcal{H}_w$ ownership matrix $(j,k) = 1$ if both products sold by same retailer/wholesaler.
- $\frac{\partial \mathbf{p^r}}{\partial \mathbf{p^w}}$ is the pass-through matrix of the retail stage, entry $(m,j) = \partial p_m^r / \partial p_j^w$. Chain rule: $-\partial\sigma_k/\partial p_j^w = \sum_m \frac{\partial p_m^r}{\partial p_j^w}\Delta_{r,(m,k)}$, so the wholesaler's matrix is the transpose of the pass-through times $\Delta_r$ (easy to get wrong in code).

Challenge: We want $\mathbf{p^r}(\mathbf{p^w})$ and $\mathbf{mc^w}$ but we only have an implicit solution for the retailer FOC. The pass-through matrix is the object that lets the upstream firm's problem be written at all.

# Pass-through Counterfactuals? (Conlon Rao 2025)

How do we solve for $\mathbf{p^w}$ under a counterfactual pass-through matrix?

- Idea: pass-through only augments the matrix $\Delta_r(\mathbf{p^r})$: $\Omega_w \equiv \mathcal{H}_w \odot (\mathcal{P}'\Delta_r)$ with $\mathcal{P} \equiv \frac{\partial \mathbf{p^r}}{\partial \mathbf{p^w}}$.
- Example: a constant retail markup or sales tax, $\mathcal{P} = \text{diag}(1+\tau_r)$, so $\mathcal{P}' = \mathcal{P}$ and $\mathcal{P}'\Delta_r = \mathcal{P}\Lambda - \mathcal{P}\Gamma$ with $\mathcal{P}\Lambda$ still diagonal.

$$\mathbf{p^w} = \mathbf{mc^w} + \Omega_w^{-1}\boldsymbol{\sigma}(\mathbf{p^r})$$

Adapt the Morrow Skerlos $\zeta$ fixed point (same sign convention as the pricing lecture, $\alpha_i > 0$). For diagonal $\mathcal{P}$, $\mathcal{H}_w \odot (\mathcal{P}\Gamma) = \mathcal{P}(\mathcal{H}_w \odot \Gamma)$, so $\mathcal{P}$ cancels out of the $\Gamma$ term and survives only on the share term:

$$\mathbf{p^w} \leftrightarrow \mathbf{mc^w} + \boldsymbol{\zeta}(\mathbf{p^w}) \quad \text{where} \quad \boldsymbol{\zeta}(\mathbf{p^w}) = \Lambda^{-1}\left[\mathcal{H}_w \odot \Gamma\right](\mathbf{p^w} - \mathbf{mc^w}) + \Lambda^{-1}\mathcal{P}^{-1}\boldsymbol{\sigma}$$

with $\Lambda, \Gamma, \boldsymbol{\sigma}$ evaluated at $\mathbf{p^r}(\mathbf{p^w})$. Higher retail pass-through (bigger $\mathcal{P}$) shrinks the wholesale markup: the wholesaler's own price moves retail prices more than one-for-one.

- General (non-diagonal) $\mathcal{P}$: $\mathcal{P}'\Lambda$ is no longer diagonal, so the clean split fails. Split instead on $D = \text{diag}(\Omega_w)$, iterate $\mathbf{p^w} - \mathbf{mc^w} \leftarrow\!\shortmid D^{-1}(D - \Omega_w)(\mathbf{p^w} - \mathbf{mc^w}) + D^{-1}\boldsymbol{\sigma}$ with no convergence guarantee, or use Newton with the pass-through Jacobian.

# What's the point? Vertical settings

Why do we care about pass-through in vertical settings?

- The wholesaler's effective demand derivative is $\mathcal{P}'\Delta_r$: with low retail pass-through the manufacturer faces a flatter response and takes a bigger margin; with high pass-through the retailer is doing the marking up.
- When retail pass-through is high we think that downstream firms have significant market power.
- This also means the scope for elimination of double marginalization is large.
  - Higher pass-through should be associated with more efficient vertical mergers? (Not sure this has been tested)
- A huge class of models (post-and-hold, minimum markups, RPM, vertical contracts in the second IO course) adds an extra parameter between wholesale and retail prices. All of them are a statement about $\mathcal{P}$.

# Taxes and the data

# Specific vs. Ad Valorem Taxes: The $\zeta$ Map Already Told Us

Producer price $\mathbf{p}$, consumer price $\mathbf{p^c}$. A specific tax $t$ adds to cost; an ad valorem tax $\tau$ scales the consumer price, $\mathbf{p^c} = (1+\tau)\mathbf{p}$, i.e. $\mathcal{P} = (1+\tau)\mathbb{I}_J$ on the previous slide:

$$\text{specific:} \quad \mathbf{p} - \mathbf{mc} - t\mathbf{1} = \Lambda^{-1}[\mathcal{H} \odot \Gamma](\mathbf{p} - \mathbf{mc} - t\mathbf{1}) + \Lambda^{-1}\boldsymbol{\sigma}$$

$$\text{ad valorem:} \quad \mathbf{p} - \mathbf{mc} = \Lambda^{-1}[\mathcal{H} \odot \Gamma](\mathbf{p} - \mathbf{mc}) + \frac{1}{1+\tau}\Lambda^{-1}\boldsymbol{\sigma}$$

- A specific tax leaves the markup equation alone: the firm re-optimizes against a higher cost and pass-through is $\rho$ from before.
- An ad valorem tax taxes the markup: the share term is scaled by $1/(1+\tau)$, so the producer gives up markup before any demand response. For the same revenue, ad valorem gives lower consumer prices under market power (Delipalla Keen 1992; Anderson, de Palma, Kreider 2001 for logit oligopoly).
- Single-product logit, rivals fixed: specific tax $\rho = 1 - \sigma_j$; ad valorem consumer-price pass-through is smaller still.
- US excise taxes (gasoline, cigarettes, alcohol, the Seattle soda tax) are specific; sales taxes are ad valorem. Conlon Rao (2025) compare volume, sales and ethanol taxes on spirits this way.

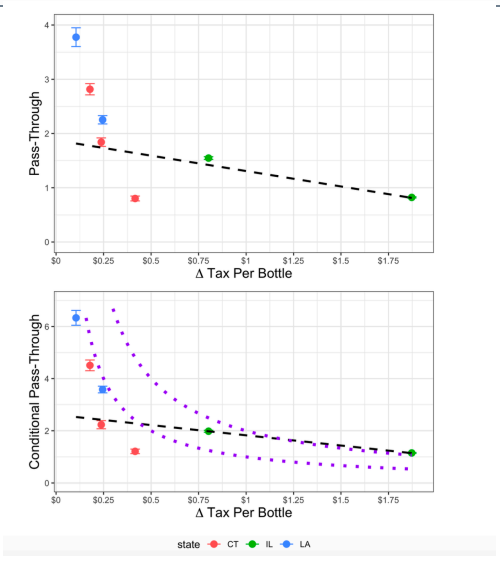
# A simple empirical exercise (Conlon Rao, AEJ:EP 2020)

- US states tax distilled spirits by volume (a specific tax)
- We see tax changes in three states (Connecticut, Louisiana, Illinois)
- Three popular bottle sizes (750mL, 1L, 1.75L)
- Run a regression at a 3 month difference product by product

$$\Delta p_{jst} = \rho_{jst}(\mathbf{X}, \Delta\tau) \cdot \Delta\tau_{jt} + \beta \Delta x_{jst} + \gamma_j + \gamma_t + \epsilon_{jst}$$

- Only innovation: allow $\rho_{jst}(\mathbf{X}, \Delta\tau)$ to vary by size of tax increase.
- Prediction from the theory so far: something below one for a logit-like product, around one with constant markups, above one only with very convex demand.

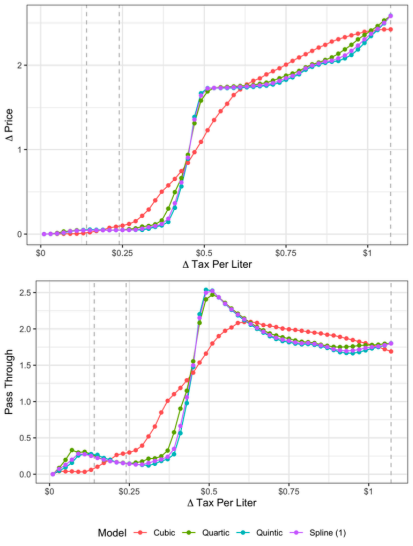
# Goes horribly wrong




- Regression coefficients are huge $> 1$ and all over the map
- Conditional on a price change even larger $> 3$
- Seem to coincide with the purple line

# Is PT a structural parameter?

- Price points are a big problem here
- Estimated an ordered probit with flexible $\beta(\Delta\tau)$
- Pass-through depends on where you are; not stable at all (!)
- Maybe structural interpretations of pass-through parameter are a bad idea!
- The theory is about a smooth FOC. Retail prices move in discrete steps ($X.99), so a 30 cent tax either does nothing or moves the price a dollar. Averaged over products that can look like anything.



# Is It Chain Pricing? DellaVigna Gentzkow (QJE 2019), Butters Sacks Seo (AER 2022)

- DellaVigna and Gentzkow (2019): US retail chains set nearly uniform prices across their stores despite large differences in local demand and income. So a natural excuse for strange local tax pass-through: the chain prices nationally, the local tax is absorbed or spread across all stores.
- Butters, Sacks, Seo (2022) test it directly: 68 local excise tax changes (beer, liquor, cigarettes, soda) and 76 sales tax changes, 35,151 stores in 96 multi-state chains.
  - Local cost shocks are passed through to local prices, at similar rates for national and local chains, and about one-for-one in levels (Sangani's benchmark).
  - No spillovers to unaffected stores in the same chain.
- So chains price uniformly with respect to demand but respond locally to costs. Uniform pricing is not the explanation for the spirits results; price points and the discreteness of retail pricing are.
- Practical upshots: (a) local tax changes are clean cost shifters for the supply side of a demand model (the Conlon Rao supply moments); (b) the elasticity you back out from a tax change is the elasticity of the stores that actually changed price, with the price-point weighting from the previous slide.

# Some challenges

- Pless and van Benthem (AEJ:Applied 2019): use pass-through $> 1$ as a test for market power? Only if you are willing to rule out convex demand under competition, and to assume constant marginal cost. The converse fails too: $\rho = 1$ in levels is not evidence of competition. Shift-invariant demand delivers it at any markup (Sangani 2026).
- Chetty, Looney, Kroft (AER 2009)

$$\log q\left(p, \tau^S\right) = \alpha + \beta \log p + \theta_\tau \beta \log\left(1 + \tau^S\right)$$
$$\theta_\tau = \frac{\partial \log q}{\partial \log\left(1 + \tau^S\right)} \Big/ \frac{\partial \log q}{\partial \log p} = \frac{\varepsilon_{q,1+\tau^s}}{\varepsilon_{q,p}}$$

- Use $\theta$ as evidence of "tax salience" (is it?). With imperfect competition the posted price responds to the tax too, so the incidence and excess-burden formulas change: Kroft, Laliberté, Leal-Vizcaíno, Notowidigdo (ReStud 2024) derive them with salience and market power together, find near-complete pass-through of sales taxes, consumers who respond more to prices than to taxes, and a marginal excess burden larger than the standard formulas give.
- Can we identify markups from pass-through alone...?!? Only jointly with curvature. Which is where we started.